

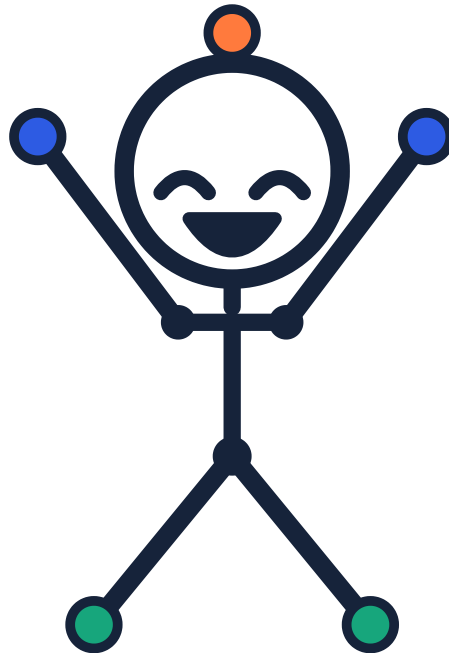


Young Innovators
FOR CHANGE
innovateyouth.org

A FULL-YEAR ENTERPRISE PROGRAM · GRADE 4

The Planning Year

36 weekly assignments. Money to the cent, a plan you wrote,
and the record that tells you whether it was any good.



This book belongs to

How long my plan runs

It starts on

Teacher

School year

Free to print, copy and use. No license, no account, no cost to any student or school.

Decide it before.

Two pages a week, front and back. About thirty-five minutes.

The shape of a week

Page 1 teaches one idea and gives you something to do with it. **Page 2** is practice, a question for home, and a harder challenge. No computer needed.

Weeks 1 to 12 are money to the cent and then the first plan. Weeks 13 to 24 are about keeping it, which means a second column saying what really happened. Weeks 25 to 36 are the money coming in, the cost nobody planned for, and a plan that runs in months.

The boxes on the page

IN CLASS

The Grade 4 standard your class is most likely on that week. This book sits beside your math lesson, not against it.

MONEY WARM-UP

Two minutes of arithmetic with money in it. Decimals to the cent, multiplying and dividing, which is what 4.NF, 4.NBT and 4.MD are for.

WORDS TO KEEP

One or two words pinned down properly. Adults use these loosely; you will not have to.

PLAN BOOK

The one thing you build all year. 20 pages by week 36, every one of them about your own money.

FROM YOUR BOOK

Which earlier pages this week needs. Week 32 is impossible without four weeks of week 15.

TALK ABOUT IT

Two questions for someone at home. Five minutes, and the part that makes the week stick.

STUCK? TRY THIS

A way in when the page stops making sense.

CHALLENGE ZONE

Harder, and usually open. For whoever finishes early.

Three things this book does not do

Nothing is sold. No product, no customer, no market day. That is The Venture Year, also Grade 4 and a different book.

Nobody is asked what anything costs at their house. Every amount is the student's own or is printed on the page.

No plan is marked right or wrong. What can be marked is whether it adds up.

WEEK	TITLE	STRAND	WHAT IT SITS BESIDE IN CLASS
TRIMESTER 1 · WEEKS 1–12 · TO THE CENT, AND THE FIRST PLAN			
1	A Dollar Has a Hundred Parts	Cents	Decimal notation for fractions (4.NF.6)
2	Which Is More?	Cents	Comparing decimals to hundredths (4.NF.7)
3	Adding to the Cent	Cents	Money problems with decimals (4.MD.2)
4	Taking Away to the Cent	Cents	Money problems with decimals (4.MD.2)
5	A Plan Is Choices Made Early	The Plan	Multi-step word problems (4.OA.3)
6	Four Weeks, One Amount	The Plan	Dividing with remainders (4.NBT.6, 4.OA.3)
7	What Goes in a Plan	The Plan	Multi-step word problems (4.OA.3)
8	The Plan That Adds Up	The Plan	Multi-step word problems (4.OA.3)
9	Price Per One	Comparing	Dividing with remainders (4.NBT.6)
10	The Same Money, Two Sizes	Comparing	Multiplying and dividing (4.NBT.5, 4.NBT.6)
11	Cheaper Is Not Always Less	Comparing	Multi-step word problems (4.OA.3)
12	The Plan Detective	Review	Multi-step word problems (4.OA.3)
TRIMESTER 2 · WEEKS 13–24 · KEEPING IT			
13	Write the Plan Down	Keeping Track	Money problems with decimals (4.MD.2)
14	Plan and Actual, Side by Side	Keeping Track	Money problems with decimals (4.MD.2)
15	The Difference Column	Keeping Track	Adding and subtracting multi-digit (4.NBT.4)
16	A Week That Went Over	Keeping Track	Multi-step word problems (4.OA.3)
17	Parts of a Whole Amount	Sharing Out	Equivalent fractions (4.NF.1)
18	Halves, Quarters, Tenths	Sharing Out	Comparing and equivalent fractions (4.NF.1, 4.NF.2)

WEEK	TITLE	STRAND	WHAT IT SITS BESIDE IN CLASS
19	All of It Adds to One	Sharing Out	Adding fractions with like denominators (4.NF.3)
20	Every Week, All Year	Repeating	Multiplying multi-digit numbers (4.NBT.5)
21	Every Month	Repeating	Multiplying multi-digit numbers (4.NBT.5)
22	Small and Often	Repeating	Multi-step word problems (4.OA.3)
23	The Thing You Did Not Plan For	The Unplanned	Multi-step word problems (4.OA.3)
24	Your Plan, Rewritten	Review	Multi-step word problems (4.OA.3)
TRIMESTER 3 · WEEKS 25–36 · COMING IN, AND GOING LONG			
25	Money Coming In	Coming In	Multi-step word problems (4.OA.3)
26	When It Is Not the Same Every Time	Coming In	Multi-step word problems (4.OA.3)
27	A Rate, and What It Buys	Coming In	Multiplying multi-digit numbers (4.NBT.5)
28	Set Some Aside First	The Unplanned	Multi-step word problems (4.OA.3)
29	How Big Should It Be?	The Unplanned	Multi-step word problems (4.OA.3)
30	A Whole Month of Both Columns	Keeping Track	Adding and subtracting multi-digit (4.NBT.4)
31	Where Did It Go?	Keeping Track	Making a line plot, reading data (4.MD.4)
32	Was the Plan Right?	Comparing	Multi-step word problems (4.OA.3)
33	Change One Thing	The Plan	Multi-step word problems (4.OA.3)
34	Months, Not Weeks	The Plan	Multi-step word problems, remainders (4.OA.3, 4.NBT.6)
35	Your Plan Book, Finished	Review	Multi-step word problems (4.OA.3)
36	Decide It Before	Review	Multi-step word problems (4.OA.3)

A Dollar Has a Hundred Parts

Cents are not a second number written after the dollars. They are pieces of the dollar.

IN CLASS THIS WEEK

Decimal notation for fractions (4.NF.6)

MONEY WARM-UP • 2 MINUTES

Write forty cents as part of a dollar. _____

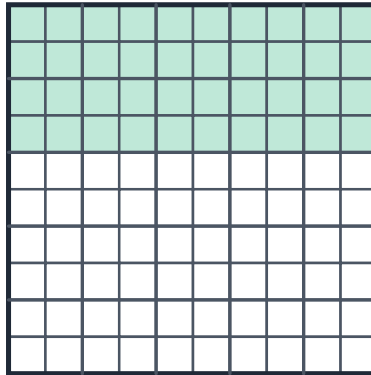
WORDS TO KEEP

cents

The hundred equal parts of a dollar. \$1.00 is one hundred of them.

decimal point

The dot that separates whole dollars from cents.



40 of the 100 parts is \$0.40

Why it is written that way

Cut a dollar into a hundred equal parts. One of them is a cent.

So \$0.40 is not a four and a zero sitting next to a dollar sign. It is **forty of the hundred parts**, which is why you can also write it 40/100, and why four dimes is the same money as forty pennies.

The dot is there to say where the whole dollars stop. Everything to the left is dollars you have. Everything to the right is pieces of one.

Write it both ways

	In words	As a fraction of a dollar	With a dollar sign
a	forty cents	40/100	\$0.40
b	seven cents		
c	ninety cents		
d	three cents		
e	sixty-five cents		

One of these is the one people get wrong most often. Which, and what do they write instead?

► ADD TO YOUR PLAN BOOK

Start your Plan Book with this page: a dollar drawn as a hundred squares, with \$0.40 shaded and written three ways.

TALK ABOUT IT

Find a price at home with cents in it. Say it out loud as parts of a dollar.

Why does a dollar have a hundred parts and not ten?

**STUCK? TRY THIS**

Say the number of cents first. Then ask yourself how many hundredths that is.

★ CHALLENGE ZONE

A dollar has 100 parts. How many parts is \$2.30? Now how many parts is \$2.03, and how far apart are those two answers?

Which Is More?

\$4.09 and \$4.90 look almost the same and are eighty-one cents apart.

IN CLASS THIS WEEK

Comparing decimals to hundredths (4.NF.7)

MONEY WARM-UP • 2 MINUTES

Which is more, \$3.5 or \$3.45? _____

WORDS TO KEEP

more

Bigger, once you have lined the decimal points up. \$4.90 beats \$4.09.



Line up the points, not the ends

Compare the dollars first. If they are the same, compare the tenths. Only then the hundredths.

\$4.09 and \$4.90 both have four dollars. Then one has zero tenths and the other has nine. That settles it before you ever look at the last digit.

The trap is reading the cents as an ordinary number and thinking 09 and 90 are nine and ninety of the same thing. They are not. Nine hundredths and ninety hundredths. One of them is nearly a whole dollar.

More, less, or the same?

	This	or this	Which is more?	By how much?
a	\$4.09	\$4.90	\$4.90	\$0.81
b	\$2.50	\$2.05		
c	\$7.30	\$7.3		
d	\$0.99	\$1.01		
e	\$6.40	\$6.04		

One row is a trick: the two are the same money written two ways. Which row, and why is it allowed?

 TALK ABOUT IT

Which is more, ninety cents or nine cents? Now say them as \$0.90 and \$0.09.

Where in a shop would getting this wrong cost you the most?

**STUCK? TRY THIS**

Write both amounts one above the other with the points in a line. The columns will answer it for you.

★ CHALLENGE ZONE

Write four different amounts that are all more than \$5.09 and less than \$5.90. How many are there altogether?

Adding to the Cent

Two prices, added properly, with nothing rounded away.

IN CLASS THIS WEEK

Money problems with decimals (4.MD.2)

MONEY WARM-UP • 2 MINUTES

$\$2.45 + \$1.30 = \underline{\hspace{2cm}}$

WORDS TO KEEP

more

Bigger, once you have lined the decimal points up. \$4.90 beats \$4.09.

RECEIPT	
notebook	\$2.45
pen	\$1.30
eraser	\$0.75

The points go under each other

$\$2.45 + \$1.30 + \$0.75$. Stack them so every point is in one vertical line, then add each column.

Ten hundredths make a tenth and ten tenths make a dollar, the same way ten ones make a ten. So the carrying works exactly as it always did. Nothing new is happening; the columns just have names now.

Write the point in the answer **before** you add, straight down from the others. Putting it in afterwards is where the lost dollar comes from.

Add these baskets

	Things in the basket	Total
a	\$2.45, \$1.30, \$0.75	\$4.50
b	\$3.20, \$0.95	
c	\$1.05, \$1.05, \$1.05	
d	\$6.40, \$2.60	
e	\$0.85, \$0.85, \$2.30	

Two of these come to a whole number of dollars with no cents left. Which two?

► ADD TO YOUR PLAN BOOK

Add a page headed To the Cent, with your three real prices and their total, added in columns.

TALK ABOUT IT

Add up three prices from a real shelf at home. Check it twice.

Why do the points have to be lined up and not the last digits?

**STUCK? TRY THIS**

Put a zero on the end of any amount with only one digit after the point. \$3.5 becomes \$3.50 and then everything lines up.

★ CHALLENGE ZONE

A basket comes to \$9.99. Add one more thing and make the total a whole number of dollars. What are the three cheapest things that would do it?

Taking Away to the Cent

What comes back, worked out exactly, before the person at the till tells you.

IN CLASS THIS WEEK

Money problems with decimals (4.MD.2)

MONEY WARM-UP • 2 MINUTES

You pay \$10 for something costing \$6.35. Your change? _____

WORDS TO KEEP

exact

To the cent, with nothing rounded. A price is always exact.



Ten dollars is ten dollars and no cents

\$10.00 – \$6.35. Write the ten as **\$10.00** and the subtraction is ordinary.

The answer is \$3.65. You can check it the other way round without a pencil: count up from \$6.35 to \$7.00, which is 65 cents, then \$7.00 to \$10.00, which is \$3. Sixty-five cents and three dollars.

Two ways to the same answer is not showing off. It is how you find out you were wrong, and it is the only checking method that works while somebody is waiting for you.

What comes back?

	It costs	You hand over	Change	Check by counting up
a	\$6.35	\$10.00	\$3.65	65c then \$3
b	\$4.80	\$5.00		
c	\$6.25	\$20.00		
d	\$7.99	\$10.00		
e	\$3.06	\$5.00		

In one row the change is more than what the thing cost. Which row, and roughly how many times more?

 TALK ABOUT IT

Buy something with a grown-up and work out the change before they do.

Is it rude to check your change? What would you actually do?

**STUCK? TRY THIS**

Write the amount you handed over with two zeros after the point. Then subtract the ordinary way.

★ CHALLENGE ZONE

You get \$1.15 in change and you were given four coins. What could they have been? Find every answer you can.

A Plan Is Choices Made Early

You already know how to choose. This year you choose in advance, for four weeks at a time.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

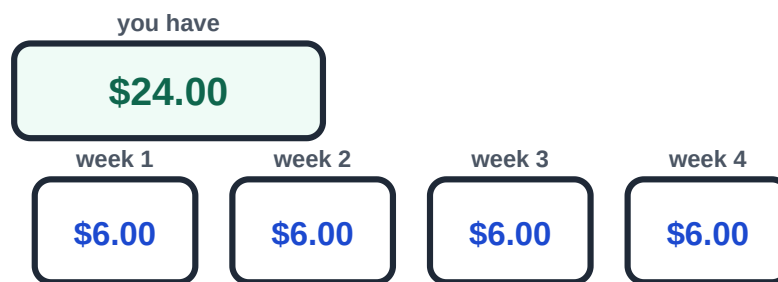
MONEY WARM-UP • 2 MINUTES

You get \$6 a week for 4 weeks. How much altogether? _____

WORDS TO KEEP

a plan

What you decided to do with your money before you were standing in front of it.



Deciding in the shop is not deciding

A plan is what you decided to do with your money before you were standing in front of the thing.

Last year you learned that picking one means not having the other. That is true whether you notice it or not. The difference a plan makes is that you do the noticing early, sitting down, when nobody is waiting and nothing is shiny.

It is not a rule somebody else made you follow. You wrote it. You are allowed to change it, and week 33 is about doing exactly that. But changing a plan on purpose is a different thing from never having had one.

Plan, or not a plan?

	What somebody said	Is it a plan?	Why
a	I will save some money	no	no amount, no time
b	I will put \$2 away every week for 4 weeks		
c	I want a \$30 game		
d	\$3 a week for snacks, \$2 a week saved		
e	I will not waste money		

The ones that are plans all have two things the others do not. What are the two things?

► ADD TO YOUR PLAN BOOK

Add a page headed My First Plan: an amount, four weeks, and what each week is for. You will come back and mark it in week 13.

TALK ABOUT IT

Tell somebody at home one plan you could make for the next four weeks.

What is the difference between a plan and a wish? Use one of the rows above.

**STUCK? TRY THIS**

A plan has an amount and a stretch of time. If you cannot point at both, it is not one yet.

★ CHALLENGE ZONE

Write a plan for \$24 over four weeks that would still work if one week you got nothing. What did you have to give up to make it survive that?

Four Weeks, One Amount

Splitting an amount evenly, and what to do with what is left over.

IN CLASS THIS WEEK

Dividing with remainders (4.NBT.6, 4.OA.3)

MONEY WARM-UP · 2 MINUTES

\$25 split evenly over 4 weeks. How much each week? _____

WORDS TO KEEP

split

Cut an amount into equal parts, one for each week.



The remainder has to go somewhere

\$25.00 into 4 weeks is \$6.25 each, and that is exact.

But \$25 into 6 weeks is \$4.16 and a bit, and the bit is real money that has to end up somewhere. You cannot hand somebody a third of a cent. So you decide: \$4.16 a week and 4 cents left at the end, or \$4.15 a week and 10 cents left, or one week gets a little more than the others.

None of those is the right answer. Choosing one and writing it down is the right answer, because the money exists whether or not your plan mentions it.

Split it

	Amount	Weeks	Each week	Left over
a	\$25.00	4	\$6.25	nothing
b	\$24.00	5		
c	\$20.00	3		
d	\$16.00	3		
e	\$22.00	6		

Two rows come out exactly even. Three do not. Of those three, one leaves more unaccounted for than the other two. Which row, and where would you put the money?

 TALK ABOUT IT

Split a real amount at home over four weeks. Is it even?

When would you rather have an uneven split than an even one?

**STUCK? TRY THIS**

Divide as usual. If it does not go in exactly, the part that will not divide is the leftover, and it is measured in cents.

★ CHALLENGE ZONE

Find an amount under \$50 that splits evenly over 4 weeks, over 5 weeks, and over 6 weeks. How did you look for it?

What Goes in a Plan

Naming the groups, so that a plan is about kinds of spending rather than about things.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP • 2 MINUTES

\$2 for snacks, \$3 for a gift, \$1 saved. Total? _____

WORDS TO KEEP

category

A group of things you spend on, named so you can add them up.

What for	Planned
snacks	\$2.00
a gift	\$3.00
saving	\$1.00
Altogether	

A category is a group with a name

A plan does not list the things. It lists the **kinds** of things, and how much each kind gets.

You cannot list the things, because you do not know yet what they will be. You do not know on Monday which snack you will want on Thursday. But you can know that snacks get two dollars.

Three or four categories is enough at this size. Ten categories for twelve dollars is not a better plan, it is a plan you will stop keeping by the second week, and a plan nobody keeps is worth nothing at all.

Sort them into categories

	The thing	Which category?
a	a chocolate bar	snacks
b	a birthday present for your cousin	
c	money put in a jar for later	
d	a comic	
e	a bag of chips shared at lunch	

Now write three categories of your own that would cover everything you spend on. Is there anything left over that fits none of them?

► ADD TO YOUR PLAN BOOK

Add a page headed My Categories, with three or four of them and one line saying what belongs in each.

TALK ABOUT IT

Ask somebody at home what categories they would use. Do they use the word?

Is a present for somebody else in the same category as a treat for you?

**STUCK? TRY THIS**

Ask what kind of thing it is, not what the thing is. A comic and a book are the same kind.

★ CHALLENGE ZONE

Two people spend exactly the same money and write completely different categories. Is one of them wrong? Say what makes a set of categories good.

The Plan That Adds Up

If the parts do not come to the whole, it is not a plan yet.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP • 2 MINUTES

\$3 + \$4 + \$2 out of \$10. How much is unassigned? _____

WORDS TO KEEP

adds up to

The parts of a plan together make the whole amount. If they do not, it is a wish.



Every dollar has a job or it does not

Add the categories up. If they come to less than what you have, the rest is not saved. It is unassigned, which is different.

Unassigned money gets spent, and it gets spent on nothing you would have chosen, because nobody chose it. That is not a character flaw. It is what happens to money with no job.

If the categories come to **more** than you have, the plan is impossible and you found out now, at a table, instead of in week three when something you promised somebody has to be canceled.

Does the plan add up?

	The categories	You have	Adds to	Too much, too little, or exact
a	\$3, \$4, \$2	\$10.00	\$9.00	\$1 not assigned
b	\$5, \$5	\$10.00		
c	\$6, \$3, \$2	\$10.00		
d	\$2.50, \$2.50, \$4	\$10.00		
e	\$4, \$4, \$4	\$10.00		

Two rows are impossible. One is exact. Two leave money with no job. Say what you would do about each of the two.

► ADD TO YOUR PLAN BOOK

Add a page headed Does It Add Up: your plan, the total of its categories, and the amount you have, with the two compared.

💬 TALK ABOUT IT

Show your plan to somebody at home and add it up in front of them.

Is money with no job the same as saved money? Argue it both ways.

**STUCK? TRY THIS**

Add the categories first, then compare that total with what you have. The comparison is the whole exercise.

★ CHALLENGE ZONE

Make a plan for \$12 with four categories where no two get the same amount and every dollar has a job. Now do it again with five categories.

Price Per One

Dividing to find what a single one costs, to the cent, instead of guessing.

IN CLASS THIS WEEK

Dividing with remainders (4.NBT.6)

MONEY WARM-UP • 2 MINUTES

6 pens cost \$3.00. What is one pen? _____

WORDS TO KEEP

per one

What a single one costs, worked out by dividing.

4 in a pack

\$3.00

each:

10 in a pack

\$6.50

each:

divide the price by how many are in it

Divide, do not eyeball it

\$3.00 for 4 is 75 cents each. \$6.50 for 10 is 65 cents each. The bigger pack is cheaper per one, and now you know it rather than assume it.

Last year you were allowed to say the big box is probably better value. It usually is.

Usually is the word doing the work there, and this year you can settle it in one division.

The answer is money, so it goes to the cent. If the division does not come out even, the price per one is between two amounts and you say so: \$1.00 for 3 is about 33 cents each, and the missing cent is real.

What does one cost?

	How many	Price	Each
a	4	\$3.00	\$0.75
b	10	\$6.50	
c	5	\$4.00	
d	8	\$2.00	
e	3	\$1.00	

One row does not divide evenly into cents. Which one, and what would you write for it?

 TALK ABOUT IT

Find two pack sizes of the same thing at home and work out each per one.

Does the shop want you to be able to do this quickly? Why might they not?

**STUCK? TRY THIS**

Divide the price by how many are in the pack. Keep the answer in cents so you can compare.

★ CHALLENGE ZONE

Two packs cost the same per one but one is a better buy anyway. Invent the two packs and explain.

The Same Money, Two Sizes

Two packs, one budget, and the arithmetic that says which one your money likes.

IN CLASS THIS WEEK

Multiplying and dividing (4.NBT.5, 4.NBT.6)

MONEY WARM-UP · 2 MINUTES

\$0.75 each. How many can you buy with \$6.00? _____

WORDS TO KEEP

per one

What a single one costs, worked out by dividing.

6 in a pack

\$4.20

each:

15 in a pack

\$9.00

each:

1 loose

\$0.80

each:

divide the price by how many are in it

Two questions, not one

How much is one? And how many do I get for what I have?

\$4.20 for 6 is 70 cents each. \$9.00 for 15 is 60 cents each. The fifteen-pack wins on price per one, comfortably.

Now the other question. If you have \$5.00, the fifteen-pack is not an option at all, and the comparison you just did does not help you. The cheapest thing per one that you cannot afford is not cheap. It is unavailable.

Work both questions

	Pack	Price	Each	How many with \$6.00
a	6	\$4.20	\$0.70	8
b	15	\$9.00		
c	1 loose	\$0.80		
d	4	\$2.00		
e	20	\$8.00		

With exactly \$6.00, which row lets you take home the most of them? Is it the same row as the cheapest per one?

**TALK ABOUT IT**

Pick something at home sold in two sizes. Which would you buy, and why?

Have you ever bought the big one and then not used it all?

**STUCK? TRY THIS**

Do the per one first, in cents. Then ask separately how many whole packs your money reaches.

★ CHALLENGE ZONE

You have \$10.00 and a thing is sold loose at \$0.80 or in fifteens at \$9.00. What is the most you can take home, and what is left in your hand?

Cheaper Is Not Always Less

The best price per one, on more than you wanted, is more money out of your hand.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP • 2 MINUTES

You need 2. A pack of 6 costs \$3.00. What do you pay? _____

WORDS TO KEEP

more than you need A price that is lower per one and higher in total, because you had to buy six.

2 loose, 90c each

\$1.80

each:

you need 2

pack of 6

\$3.00

each:

you need 2

divide the price by how many are in it

Per one is not the only number

Two loose ones at 90 cents is \$1.80. A pack of six is \$3.00, which is 50 cents each. The pack is far cheaper per one and costs you \$1.20 more today.

Both of those sentences are true at the same time and people argue about which one counts. Here is the honest answer: it depends entirely on whether you will use the other four.

If you will, the pack was cheaper. If they sit in a drawer for a year, you paid \$3.00 for two, which is \$1.50 each, and the cheapest-looking option was the most expensive one you could have picked.

What do you actually pay?

	You need	Loose price	Pack	Loose total	Pack total
a	2	\$0.90	6 for \$3.00	\$1.80	\$3.00
b	5	\$0.90	6 for \$3.00		
c	8	\$0.90	6 for \$3.00		
d	12	\$0.90	6 for \$3.00		
e	1	\$0.90	6 for \$3.00		

There is a number of them where the pack stops being worse and starts being better. Find it, and say what happens exactly at that number.

► ADD TO YOUR PLAN BOOK

Add a page headed More Than I Needed: the two options, both totals, and which one you would take.

TALK ABOUT IT

Tell somebody at home about a time buying more was the wrong call.

Who benefits from the pack being cheaper per one?

**STUCK? TRY THIS**

Work out what leaves your hand today for each option. Compare those two numbers, not the per one.

★ CHALLENGE ZONE

The pack is 6 for \$3.00 and loose are 90 cents. For which numbers needed is the pack cheaper? Write the whole list from 1 to 12.

The Plan Detective

Everything from weeks 1 to 11, on one plan somebody else wrote badly.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP • 2 MINUTES

\$4.09, \$4.90, \$1.01. Which is the largest? _____

What for	Planned
snacks	\$4.09
presents	\$4.90
other	\$2.90
comics	\$1.01
Altogether	\$12.00

Four things wrong with this plan

Somebody has \$12.00 and wrote the plan above. It has at least four problems and every one of them is something you learned since week 1.

Start with the arithmetic: does it add up? Then look at the two amounts at the top and read them carefully. Then look at what the categories are called. Then ask what a plan needs that this one never says.

Write your four in sentences, not in numbers. A detective says what happened, not just which number is wrong.

The evidence

	The question	Your answer
a	Do the four categories add up to \$12.00?	
b	If not, how far off and in which direction?	
c	Which two amounts look almost the same? How far apart are they?	
d	Which category has a name that does not say what it is for?	
e	Week 5 said a plan needs two things. Which one is missing here?	

Now rewrite the plan so it adds to exactly \$12.00, keeping the same four categories, and say what you changed and why.

**TALK ABOUT IT**

Show your rewritten plan to somebody at home and defend one change.

Which of the four problems would matter most in real life?

**STUCK? TRY THIS**

Add the four amounts first. Everything else is easier once you know whether the plan is over or under.

★ CHALLENGE ZONE

Write a plan for \$12.00 that adds up exactly, has four categories, and where no category gets a whole number of dollars.

Write the Plan Down

The plan column. Filled in before the week starts, and not touched afterwards.

IN CLASS THIS WEEK

Money problems with decimals (4.MD.2)

MONEY WARM-UP • 2 MINUTES

$\$2.50 + \$3.25 + \$1.75 = \underline{\hspace{2cm}}$

WORDS TO KEEP

the plan column What you said you would spend, written before the week starts.

What for	Planned
snacks	\$2.50
presents	\$3.25
saving	\$1.75
Altogether	\$7.50

Written before, in ink

The plan column is what you said you would spend. It is written before the week starts and then it does not change.

That last part is the whole trick. If you are allowed to edit the plan while the week is happening, then at the end of the week the plan will match perfectly, every time, and you will have learned nothing at all. A plan that is always right is not being checked.

Next week a second column arrives beside it, and the gap between them is the only thing in this book that can tell you something about yourself.

Fill in the plan column

	What for	Planned
a	snacks	\$2.50
b	presents	\$3.25
c	saving	\$1.75
d	(yours)	
e	Altogether	

Add the three. Then write your own plan for the coming week in the same shape, with your own categories and your own amounts.

► ADD TO YOUR PLAN BOOK

Add a page headed The Plan Column, with next week's plan filled in and the date you wrote it.

TALK ABOUT IT

Tell somebody at home what your plan column says, before the week starts.

Why should the plan column not be changed once the week has begun?

**STUCK? TRY THIS**

Write the categories down the left. Amounts down the right. Total at the bottom, underlined.

★ CHALLENGE ZONE

Write a plan column where the categories add to exactly \$7.50 and every amount ends in 25, 50 or 75 cents.

Plan and Actual, Side by Side

The second column, filled in as the week happens, next to the first.

IN CLASS THIS WEEK

Money problems with decimals (4.MD.2)

MONEY WARM-UP • 2 MINUTES

Planned \$2.50, spent \$3.10. More or less? _____

WORDS TO KEEP

the actual column What you really spent, written as it happens.

What for	Planned	Actual
snacks	\$2.50	\$3.10
presents	\$3.25	\$3.25
saving	\$1.75	\$1.00

Two columns, written at two different times

The plan column was written before. The actual column is written as the money leaves, on the day it leaves.

On the day, not on Sunday from memory. Memory is generous about small amounts and this book is not going to pretend otherwise. Writing \$1.20 down on Tuesday takes four seconds; reconstructing Tuesday on Sunday takes ten minutes and gets it wrong.

Notice the middle row above. Planned \$3.25, spent \$3.25. That is allowed to happen and it is worth noticing when it does.

Read the two columns

	What for	Planned	Actual	Which is bigger?
a	snacks	\$2.50	\$3.10	actual
b	presents	\$3.25	\$3.25	
c	saving	\$1.75	\$1.00	
d	comics	\$1.00	\$0.00	
e	Altogether			

Add each column. Which total is bigger, and does that match what you would have guessed from looking at the rows?

► ADD TO YOUR PLAN BOOK

Add a page headed Plan and Actual, with both columns for one real week of your own.

TALK ABOUT IT

Write down every amount you spend for three days, on the day.

Which is harder, making the plan or writing down what really happened?

**STUCK? TRY THIS**

Two totals at the bottom, one under each column. Compare the totals last, after the rows.

★ CHALLENGE ZONE

Invent a week where three rows are over plan and the total is still under plan. What has to be true?

The Difference Column

Over, under, or exactly nothing. The third column is where the plan answers back.

IN CLASS THIS WEEK

Adding and subtracting multi-digit (4.NBT.4)

MONEY WARM-UP • 2 MINUTES

Planned \$1.75, spent \$1.00. The difference? _____

WORDS TO KEEP

the difference Plan minus actual. Over, under, or exactly nothing.

What for	Planned	Actual	Over or under
snacks	\$2.50	\$3.10	\$0.60 over
presents	\$3.25	\$3.25	nothing
saving	\$1.75	\$1.00	\$0.75 under
comics	\$1.00	\$0.00	

Say which way, not just how much

Plan minus actual. But 60 cents is not an answer on its own: 60 cents **over** and 60 cents **under** are opposite pieces of news.

So every difference gets a direction written next to it. Over means you spent more than you said. Under means you spent less. Nothing means they matched, which is worth writing down rather than leaving blank, because a blank looks like you forgot.

Under is not automatically good, by the way. The saving row above is 75 cents under, and being under on saving means you saved less than you meant to.

Fill in the difference

	What for	Planned	Actual	Difference, and which way
a	snacks	\$2.50	\$3.10	\$0.60 over
b	presents	\$3.25	\$3.25	
c	saving	\$1.75	\$1.00	
d	comics	\$1.00	\$0.00	
e	bus	\$2.00	\$2.40	

Two rows are under and being under is bad news in one of them and good news in the other. Which is which, and why?

► ADD TO YOUR PLAN BOOK

Add a page headed The Difference, with all three columns for your week and a sentence about the biggest gap.

TALK ABOUT IT

Work out the difference column for your own real week.

Is being under always good? Find the row that says no.

**STUCK? TRY THIS**

Take the smaller from the bigger, then write over or under depending on which one was bigger.

★ CHALLENGE ZONE

The difference column for five categories comes to zero altogether, and not one row matched. Invent it.

A Week That Went Over

One category is empty and it is Thursday. This is the ordinary case, not the disaster.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP • 2 MINUTES

Snacks got \$3.00. You have spent \$3.40. Over by? _____

WORDS TO KEEP

ran out

A category with nothing left in it, while the week is not over.

What for	Planned	Actual	Over or under
snacks	\$3.00	\$3.40	\$0.40 over
comics	\$2.00	\$0.50	\$1.50 under
saving	\$2.00	\$2.00	nothing

Three honest things you can do

Snacks is 40 cents over with three days left. Comics is \$1.50 under. Nobody is in trouble; a decision is needed.

Move it. Take 40 cents from comics and write it in both places, so the plan still adds up. **Stop.** Snacks is done for this week and that is what running out means. **Go over.** Spend it anyway and write down that the week finished over, so next week's plan knows.

All three are real answers. The one that is not an answer is spending it and not writing it down, because then the plan quietly becomes a story about a person who does not exist.

What would you do?

	The situation	Move it, stop, or go over?	Why
a	Snacks over 40c, comics \$1.50 under		
b	Snacks over 40c, everything else exact		
c	Presents over \$2, and the present is for tomorrow		
d	Bus over 80c and you have to get home		
e	Snacks over \$1.50 on Monday		

One of these has only one honest answer. Which, and what does that tell you about categories you cannot stop?

 TALK ABOUT IT

Ask somebody at home what they do when one category runs out.

Is moving money between categories cheating? Say what makes it not cheating.

**STUCK? TRY THIS**

Ask two questions: does this have to happen, and is there another category with room in it?

★ CHALLENGE ZONE

Your plan has four categories. Two of them are ones you cannot stop. How should that change the amounts you put in them at the start?

Parts of a Whole Amount

A fraction of money, not a fraction of a shape. Half of \$12 is a number of dollars.

IN CLASS THIS WEEK

Equivalent fractions (4.NF.1)

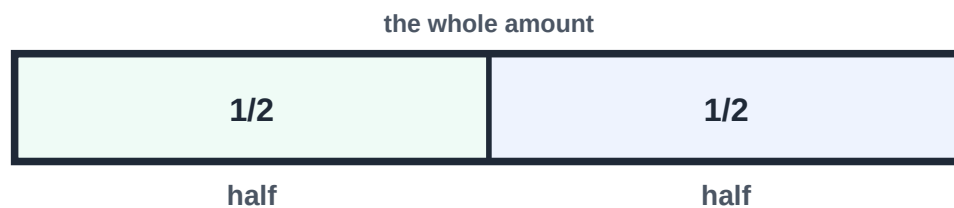
MONEY WARM-UP · 2 MINUTES

Half of \$12.00 is _____

WORDS TO KEEP

a part of it

A fraction of an amount of money, not of a shape.



Cut the amount, not the picture

Half of \$12.00 is \$6.00. A quarter of \$12.00 is \$3.00. Three quarters is \$9.00.

In maths class this year you cut shapes into equal parts and compare them. Here the thing being cut is an amount of money, and the answer comes out in dollars and cents rather than as a fraction.

To find a quarter, divide by 4. To find three quarters, divide by 4 and then multiply by 3, or take a quarter away from the whole. Both routes arrive at \$9.00, and doing it both ways is how you find out you did it right.

Take the part

	The amount	The part	How much
a	\$12.00	one half	\$6.00
b	\$12.00	one quarter	
c	\$12.00	three quarters	
d	\$20.00	one tenth	
e	\$20.00	three tenths	

Which is more, three quarters of \$12.00 or one half of \$20.00? Show the working, not just the answer.

 TALK ABOUT IT

Take a real amount at home and find a half and a quarter of it.

Why is half of a big amount more than half of a small one, even though both are halves?

**STUCK? TRY THIS**

Divide by the bottom number of the fraction. Then multiply by the top one.

★ CHALLENGE ZONE

Find an amount of money where one third of it and one quarter of it are both whole numbers of dollars with no cents. What is the smallest one?

Halves, Quarters, Tenths

Two fractions that look different and come to the same money.

IN CLASS THIS WEEK

Comparing and equivalent fractions (4.NF.1, 4.NF.2)

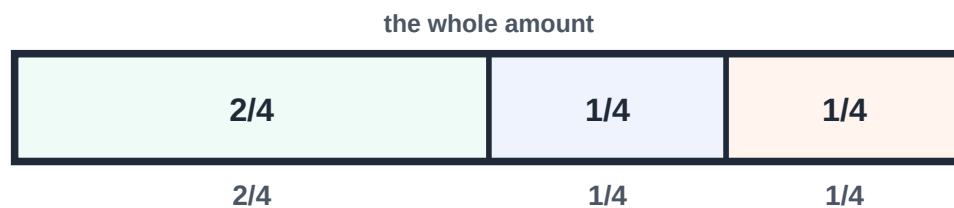
MONEY WARM-UP • 2 MINUTES

Which is more of \$20, one half or six tenths? _____

WORDS TO KEEP

the same size

Two fractions that come to the same money. A half and two quarters.



Same money, different words

Half of \$20.00 is \$10.00. Five tenths of \$20.00 is \$10.00. Two quarters of \$20.00 is \$10.00. Same money, three ways of saying it.

This matters because people describe plans out loud in fractions and they do not all use the same ones. One person says half goes to saving, another says five tenths, and they mean the identical thing.

The way to settle any of it is to turn the fraction into money. Money does not have opinions: \$10.00 is \$10.00 however you arrived at it.

Same or different?

	Of \$20.00	and	Same money?	How much each
a	one half	five tenths	yes	\$10.00 and \$10.00
b	one quarter	two tenths		
c	three quarters	seven tenths		
d	one half	two quarters		
e	one tenth	one quarter		

Two rows come out the same. In the three that do not, which of the pair is bigger, and by how much?

► ADD TO YOUR PLAN BOOK

Add a page headed Same Money, Different Words, with your plan written once in dollars and once in fractions.

TALK ABOUT IT

Say your own plan out loud in fractions instead of amounts.

Is it easier to think in fractions or in dollars? Does it depend on the amount?

**STUCK? TRY THIS**

Turn both into money first. Then the comparison is just two amounts.

★ CHALLENGE ZONE

For which amounts of money do one quarter and two tenths come out to a whole number of cents for both? Try \$20, \$25 and \$30.

All of It Adds to One

Spend, save and give, written as fractions that have to reach one whole.

IN CLASS THIS WEEK

Adding fractions with like denominators (4.NF.3)

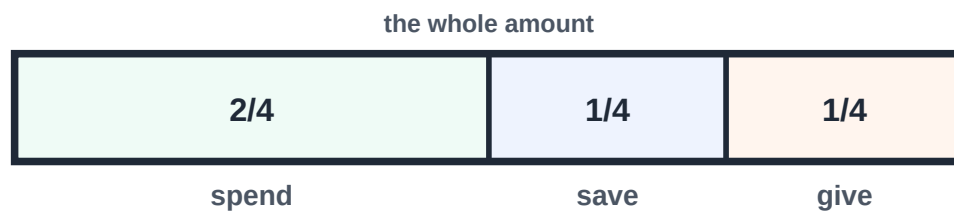
MONEY WARM-UP • 2 MINUTES

$$1/4 + 2/4 = \underline{\hspace{2cm}}$$

WORDS TO KEEP

all of it

The three parts of a plan add to one whole. Nothing is left unassigned.



Quarters that reach four quarters

Two quarters to spend, one quarter to save, one quarter to give. $2/4 + 1/4 + 1/4 = 4/4$, which is one whole.

This is week 8 again, said differently. There the categories had to add to the amount you had. Here the fractions have to add to one. They are the same requirement, and if you can see that, you can check any plan in whichever form somebody hands it to you.

Adding fractions with the same bottom number is easy: add the tops, keep the bottom. The bottom is what you cut the money into, and cutting does not change just because you are counting the pieces.

Does it reach one whole?

	Spend	Save	Give	Adds to	One whole?
a	$\frac{2}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{4}{4}$	yes
b	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$		
c	$\frac{3}{5}$	$\frac{1}{5}$	$\frac{1}{5}$		
d	$\frac{5}{10}$	$\frac{2}{10}$	$\frac{2}{10}$		
e	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{3}$		

One row goes past one whole. What would that mean if it were a real plan, and which of the three parts would you cut?

► ADD TO YOUR PLAN BOOK

Add a page headed Spend, Save, Give, with your three fractions, your three amounts, and the check that they reach one whole.

TALK ABOUT IT

Write your own plan as three fractions. Do they reach one?

Is it possible for a plan to add to less than one whole on purpose?

**STUCK? TRY THIS**

Add the top numbers. If the bottom numbers are not the same, change one of them first so they are.

★ CHALLENGE ZONE

Write a spend, save and give plan using tenths where no part is the same size and the give part is the smallest. How many different ones can you make?

Every Week, All Year

A number too small to care about once, multiplied by the times it comes back.

IN CLASS THIS WEEK

Multiplying multi-digit numbers (4.NBT.5)

MONEY WARM-UP • 2 MINUTES

\$3.00 a week for 10 weeks is _____

WORDS TO KEEP

every week

A cost that comes back. Multiply it by the weeks to see it properly.

\$3.00 every week



13 of them

Multiply it before you decide

\$3.00 a week does not sound like a decision. Over a school year of 40 weeks it is \$120.00, which absolutely is one.

Nobody is saying do not spend it. The point is that the number you should be looking at when you decide is \$120.00, and the number in front of you is \$3.00. Those feel like different sizes of question and they are the same question.

This is the reason a plan covers a stretch of time rather than a day. A day cannot show you a repeating cost, because on any single day the repeating cost looks like three dollars.

What does it come to?

	Each week	Weeks	Altogether
a	\$3.00	10	\$30.00
b	\$3.00	40	
c	\$1.50	40	
d	\$0.75	40	
e	\$5.00	52	

One row is a cost of less than a dollar and still comes to more than a week of somebody's plan. Which, and what does that tell you?

 TALK ABOUT IT

Find one thing at home that gets bought every week. Multiply it out.

Would knowing the year total change what your family buys? Should it?

**STUCK? TRY THIS**

Multiply the weekly amount by the number of weeks. Keep the cents in until the end.

★ CHALLENGE ZONE

A thing costs \$2.50 a week. How many weeks before it has cost more than a \$60 thing you could have bought once?

Every Month

Twelve times a year instead of fifty-two, and it still adds up faster than it feels.

IN CLASS THIS WEEK

Multiplying multi-digit numbers (4.NBT.5)

MONEY WARM-UP • 2 MINUTES

\$8.00 a month for 12 months is _____

WORDS TO KEEP

every month

The same, twelve times a year instead of fifty-two.

\$8.00 every month



12 of them

A month hides it better than a week

\$8.00 a month is \$96.00 a year. Twelve is a small number and it does most of its work out of sight.

Monthly costs are harder to notice than weekly ones, because a month is long enough that you have forgotten paying it last time. That is not an accident in how things are priced.

The arithmetic is not the hard part here. Multiplying by 12 is something you can already do. The hard part is remembering to, for a cost that nobody is putting in front of you.

A year of it

	Each month	A year is	More or less than \$100?
a	\$8.00	\$96.00	less
b	\$10.00		
c	\$4.50		
d	\$12.99		
e	\$2.25		

Put the five monthly amounts in order from smallest to largest. Now put the five yearly amounts in order. Is it the same order? Why must it be?

 TALK ABOUT IT

Ask somebody at home about one thing paid every month. Do not ask the amount. Why might a shop prefer to charge you monthly rather than once?

**STUCK? TRY THIS**

Multiply by 12. Then compare with \$100 by lining the numbers up.

★ CHALLENGE ZONE

One cost is \$9.00 a month and another is \$2.00 a week. Which is more over a year, and by how much? What made this harder than it looked?

Small and Often

The costs that are too small to argue about and large enough to matter.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP · 2 MINUTES

\$0.99 three times a week for 4 weeks is _____

WORDS TO KEEP

small and often A cost too small to notice once and large enough to matter added up.

\$0.99 every time



12 of them

Twice is not a habit and forty times is

Ninety-nine cents, three times a week, for a school year: $\$0.99 \times 3 \times 40 = \118.80 .

Every single one of those was an easy yes. That is the whole mechanism. No individual ninety-nine cents was ever worth stopping to think about, and the total is a bicycle.

This is not an argument for never buying anything. It is an argument for knowing the yearly number of the things you buy often, because that is the only number that lets you compare them against the things you buy rarely.

How often, and how much

	Each time	Times a week	Over 40 weeks
a	\$0.99	3	\$118.80
b	\$0.50	5	
c	\$2.00	1	
d	\$1.25	2	
e	\$0.25	10	

Row e is a quarter, which is the smallest amount on this page, and it is not the smallest total. Put the five totals in order and say what decides the order.

► ADD TO YOUR PLAN BOOK

Add a page headed Small and Often: one thing you buy often, the amount, how often, and the 40-week total.

TALK ABOUT IT

Count how many times in one week you spend under a dollar. Just count.

Is a small cost that repeats worse than one big cost of the same size?

**STUCK? TRY THIS**

Multiply the amount by the times a week first. Then multiply that by the weeks.

★ CHALLENGE ZONE

Two costs come to the same over 40 weeks. One is under a dollar and happens often, the other is over five dollars and happens rarely. Invent them both.

The Thing You Did Not Plan For

A cost nobody could have known about. It is not a mistake and the plan still has to eat it.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP · 2 MINUTES

Your plan has \$2.00 spare. A cost of \$6.50 arrives. Short by? _____

WORDS TO KEEP

did not plan for A cost you could not have known about. Not a mistake.

What for	Planned	Actual	Over or under
snacks	\$3.00	\$3.00	nothing
presents	\$4.00	\$4.00	nothing
saving	\$2.00	\$0.00	\$2.00 under
the broken thing	\$0.00	\$6.50	\$6.50 over

Every plan meets one of these

Something breaks. Something gets lost. A friend's birthday arrives on a date you did not check. The plan said nothing about it because the plan could not have.

Look at the bottom row above. The plan column says \$0.00, which is honest: nothing was planned. The actual column says \$6.50, which is also honest. Writing the row in with a zero is better than leaving it out, because leaving it out makes the month look like a month that did not happen.

Where does the \$6.50 come from? Somewhere. Saving got emptied above and it is still \$4.50 short. Week 28 is about arranging, in advance, for there to be a better answer than emptying saving.

Where does it come from?

	What happened	It costs	Spare in the plan	Short by
a	shoes came apart	\$6.50	\$2.00	\$4.50
b	lost a library book	\$8.00	\$3.50	
c	a birthday you forgot	\$5.00	\$5.00	
d	bus fare went up	\$1.20	\$0.00	
e	broke somebody's thing	\$12.00	\$4.00	

One of these five is not really unplanned, and a better plan would have had it in. Which one, and which category would it have gone in?

TALK ABOUT IT

Ask somebody at home about a cost that arrived with no warning.

Is a birthday you forgot the same kind of surprise as a shoe coming apart?

**STUCK? TRY THIS**

Take the spare away from the cost. What is left is what has to come from somewhere else.

★ CHALLENGE ZONE

Over a year, how many surprises like these would you guess arrive? What would you have to set aside each week to cover that many?

Your Plan, Rewritten

Everything from weeks 13 to 23, applied to your own plan, on purpose.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP • 2 MINUTES

A plan of \$3, \$4, \$2 and a surprise of \$5. Total needed? _____

What for	Planned	Actual	Over or under

Rewrite it with what you know now

Your first plan was written in week 5, before you had kept one. You have kept one for ten weeks. Write the second one.

Look back at your difference columns. Which category was over nearly every week? That amount was wrong, not your behavior, and the fix is to put more in it and less somewhere else. Which category was under every week? That money was never going to be spent there.

Change the amounts. Keep the categories unless one of them never had anything in it. And say, in a sentence under the plan, what you changed and what evidence made you change it.

The evidence for the new plan

	Question	Your answer
a	Which category was over most often?	
b	By about how much, in a normal week?	
c	Which category was under most often?	
d	What is the new amount for each of those two?	
e	Does the new plan still add up to what you have?	

If you moved money from one category to another, the total did not change. Check that it did not, and write the two totals side by side.

► ADD TO YOUR PLAN BOOK

Add a page headed Plan Two, with the new plan, the old plan beside it, and one sentence of evidence for every change.

TALK ABOUT IT

Show both plans, old and new, to somebody at home and say what changed.

Is changing the plan admitting the first one was bad?

**STUCK? TRY THIS**

Read your own difference column before you write anything. The new plan comes out of it, not out of your head.

★ CHALLENGE ZONE

Write a plan you are confident you can keep for four weeks with no changes. What did you have to do to be that confident, and did it cost you anything?

Money Coming In

A plan has two sides. Everything so far has been the going-out one.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

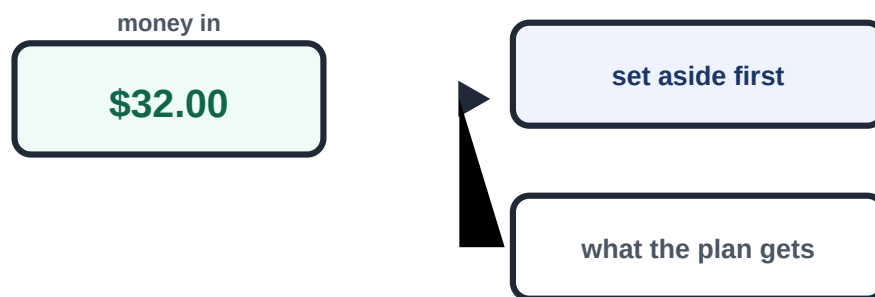
MONEY WARM-UP · 2 MINUTES

\$5 a week pocket money, \$12 from a job. Four weeks? _____

WORDS TO KEEP

coming in

Money arriving. A plan needs this side as well as the spending side.



You cannot plan what you have not counted

Coming in over four weeks: \$5.00 a week is \$20.00, plus \$12.00 once, is \$32.00.

Until now every plan in this book started from an amount somebody handed you at the top of the page. Real plans start by working that amount out, and it usually comes from more than one place.

Two kinds arrive: the amount that turns up every week whether or not you do anything, and the amount that turns up because you did something. Both are money. They behave differently in a plan and week 26 is about that.

What is coming in?

	Every week	One-offs	Over 4 weeks
a	\$5.00	\$12.00	\$32.00
b	\$3.00	\$0.00	
c	\$2.50	\$20.00	
d	\$0.00	\$40.00	
e	\$7.00	\$5.00	

Two rows come to the same over four weeks and would need very different plans. Which two, and what is different about planning them?

► ADD TO YOUR PLAN BOOK

Add a page headed Coming In, with your own four-week total and where each part of it comes from.

TALK ABOUT IT

Work out everything that comes in to you over four weeks. Write the list. Which of your money arrives whether or not you do anything?

**STUCK? TRY THIS**

Multiply the weekly amount by four, then add the one-offs. Two steps, in that order.

★ CHALLENGE ZONE

Somebody gets \$40 in one lump at the start of four weeks. Somebody else gets \$10 a week. Same money. Write one advantage each of them has.

When It Is Not the Same Every Time

You cannot plan against a number that moves. So plan against the smallest one.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

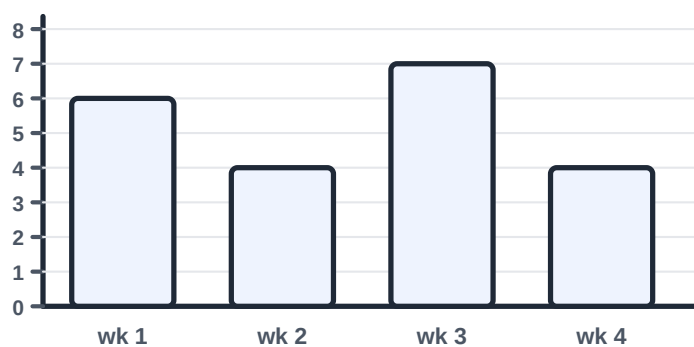
MONEY WARM-UP • 2 MINUTES

Four weeks: \$6, \$4, \$7, \$4. What is the smallest? _____

WORDS TO KEEP

varies

Different each time. You cannot plan against a number that moves, so plan against the smallest one.



Plan against the low week

Four weeks of \$6, \$4, \$7 and \$4. That is \$21.00 altogether, which is \$5.25 a week on average, and there is not a single week where you got \$5.25.

If you build a plan on \$5.25 a week, it fails in weeks 2 and 4, and it fails for a reason that has nothing to do with how carefully you spent. The average was never in the room.

Build the plan on \$4.00, the smallest. Then every week works, and the extra in a good week is a real decision you get to make rather than a hole you have to fill. That is the whole technique and adults pay for advice that amounts to it.

Which number would you plan on?

	The four weeks	Total	Average	Smallest
a	\$6, \$4, \$7, \$4	\$21.00	\$5.25	\$4.00
b	\$5, \$5, \$5, \$5			
c	\$10, \$2, \$2, \$2			
d	\$8, \$6, \$8, \$6			
e	\$3, \$9, \$3, \$9			

In one row the average and the smallest are the same number. What does that tell you about that row, and why is it the easiest to plan?

 TALK ABOUT IT

Ask somebody at home whether their money is the same amount every time.

Is planning on the smallest week too careful? What does it cost you?

**STUCK? TRY THIS**

Add them for the total. Divide by four for the average. Then just look for the smallest.

★ CHALLENGE ZONE

Invent four weeks where the average is \$6.00 and a plan built on the average would fail in three of the four weeks.

A Rate, and What It Buys

What you earn each hour, times the hours, to the cent.

IN CLASS THIS WEEK

Multiplying multi-digit numbers (4.NBT.5)

MONEY WARM-UP • 2 MINUTES

\$4.50 an hour for 3 hours is _____

WORDS TO KEEP

a rate

An amount for each one. Each hour, each job, each week.



3 hours at \$4.50

Multiply the rate, then compare it to a price

\$4.50 an hour for 3 hours is \$13.50. That is a multiplication with a decimal in it, and it works exactly like one without.

Grade 3 asked how many weeks until you could afford something. This asks a harder version: how many hours, at this rate. The answer is often not a whole number of hours, and a job does not usually let you work a third of one.

So you round up. Two and a half hours of work to afford it means three hours, because two gets you nearly there and nearly there does not buy anything.

Hours and money

	Rate	Hours	Earned	Hours to reach \$27.00
a	\$4.50	3	\$13.50	6
b	\$3.00	4		
c	\$5.40	5		
d	\$2.25	8		
e	\$6.00	2		

In one row the hours needed do not come out whole. Which, and how many hours would you actually have to work?

 TALK ABOUT IT

Work out what three hours of some real job would pay.

Is an hour of your time worth the same on a Saturday as on a school night?

**STUCK? TRY THIS**

Multiply the rate by the hours for what you earn. For the other column, divide the target by the rate.

★ CHALLENGE ZONE

Two jobs: \$5.00 an hour for 4 hours, or \$18.00 for the whole job however long it takes. When is each one better? Say what you need to know first.

Set Some Aside First

Take it off the top, before the plan is allowed to see it.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

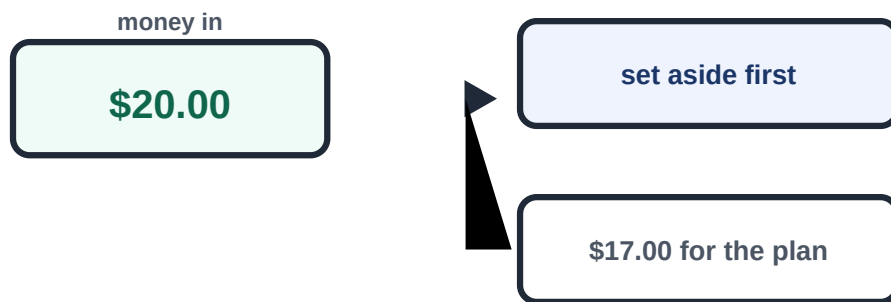
MONEY WARM-UP · 2 MINUTES

\$20.00 comes in. Set aside \$3.00. The plan gets? _____

WORDS TO KEEP

set aside

Take it out first, before the plan gets to spend it.



First, not last

\$20.00 arrives. \$3.00 goes straight into the set-aside. The plan is written for \$17.00 and never knew about the other three.

Everybody's first instinct is to set aside whatever is left at the end. Nothing is ever left at the end. This is not a fact about you, it is a fact about how spending fills whatever space it is given, and it is why week 8 said unassigned money gets spent.

So the set-aside is taken out first and the plan is built on the smaller number. The plan is tighter, which is the cost, and the cost is worth paying because week 23 is going to happen again.

Off the top

	Comes in	Set aside	The plan gets
a	\$20.00	\$3.00	\$17.00
b	\$12.00	\$1.20	
c	\$32.00	\$4.50	
d	\$8.00	\$0.80	
e	\$25.00	\$5.00	

In two rows the set-aside is a tenth of what came in. Which two? In one row it is a fifth. Which?

► ADD TO YOUR PLAN BOOK

Add a page headed Set Aside First, with what comes in, what goes aside, and what the plan is written for.

TALK ABOUT IT

Tell somebody at home the difference between saving what is left and setting aside first. Does setting aside first mean you spend less altogether, or the same?

**STUCK? TRY THIS**

Subtract the set-aside from what came in. Whatever is left is the whole amount the plan may use.

★ CHALLENGE ZONE

You want to set aside a tenth every week and your money varies: \$6, \$4, \$7, \$4. What do you set aside each week, and what does the plan get each week?

How Big Should It Be?

Sized against what a month of your plan costs, not against a feeling.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

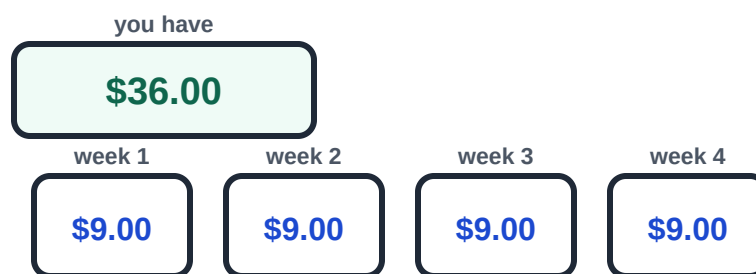
MONEY WARM-UP · 2 MINUTES

Your plan is \$9.00 a week. What does a month cost? _____

WORDS TO KEEP

how big

The size of the set-aside, decided by what a month costs.



One month of your own plan

A plan of \$9.00 a week costs \$36.00 for four weeks. A set-aside worth one month of your plan is \$36.00.

Where does that number come from? It is a rule of thumb, and a rule of thumb is a decision somebody made that turned out to work often enough to keep. The reason a month is the usual answer is that most surprises are smaller than a month of ordinary spending, so a month covers most of them.

At \$3.00 a week set aside, reaching \$36.00 takes twelve weeks. That is a long time and it is the honest answer. The set-aside is not a thing you have, it is a thing you build, and week 23 can arrive in week two.

Build the set-aside

	Plan each week	A month costs	Set aside each week	Weeks to get there
a	\$9.00	\$36.00	\$3.00	12
b	\$5.00		\$2.00	
c	\$12.00		\$4.00	
d	\$7.50		\$2.50	
e	\$10.00		\$8.00	

Row e reaches the target fastest and is the least likely to actually happen. Say why, using the words from week 8.

**TALK ABOUT IT**

Work out what a month of your own plan costs. That is your target.

Is it better to build a small set-aside quickly or a big one slowly?

**STUCK? TRY THIS**

Multiply your weekly plan by four for the target. Then divide the target by what you set aside each week.

★ CHALLENGE ZONE

You set aside \$2.00 a week and after 6 weeks something costs you \$10.00 out of it. How many weeks from the start until you reach a month of \$28.00?

A Whole Month of Both Columns

Four weeks of plan and actual, kept properly, which is harder than one.

IN CLASS THIS WEEK

Adding and subtracting multi-digit (4.NBT.4)

MONEY WARM-UP • 2 MINUTES

Four weeks at \$9.00 planned. Spent \$8, \$11, \$9, \$10. Total spent? _____

WORDS TO KEEP

the actual column What you really spent, written as it happens.

week 1	week 2	week 3	week 4
plan _____	plan _____	plan _____	plan _____
actual _____	actual _____	actual _____	actual _____

both columns, every week of the month

The month says something a week cannot

Planned \$9.00 a week, so \$36.00 for the month. Spent \$8.00, \$11.00, \$9.00 and \$10.00, which is \$38.00. Two dollars over, across a month.

Look at what the weeks alone would have told you. Week 1 under, week 2 over by two, week 3 exact, week 4 over by one. Four separate small stories.

The month is one story, and it is the true one: this plan is about two dollars a month too small. You could not have known that from any single week, and knowing it changes the plan rather than your behavior.

The month, week by week

	Week	Planned	Actual	Difference
a	1	\$9.00	\$8.00	\$1.00 under
b	2	\$9.00	\$11.00	
c	3	\$9.00	\$9.00	
d	4	\$9.00	\$10.00	
e	Month			

Add each column and fill in the Month row. Is the month difference the same as the four week differences added together? Check it.

► ADD TO YOUR PLAN BOOK

Add a page headed A Whole Month, with four weeks of both columns and the month totals underlined.

TALK ABOUT IT

Keep both columns for four weeks. Bring the month total, not the weeks.

Which is more useful to know, one week's difference or the month's?

**STUCK? TRY THIS**

Total each column down the page. The month difference is the two totals compared, and it should match the differences added up.

★ CHALLENGE ZONE

Invent a month where every single week is over plan and the month total is under. Is that possible? Say why.

Where Did It Go?

The month sorted by category, which answers a different question from the total.

IN CLASS THIS WEEK

Making a line plot, reading data (4.MD.4)

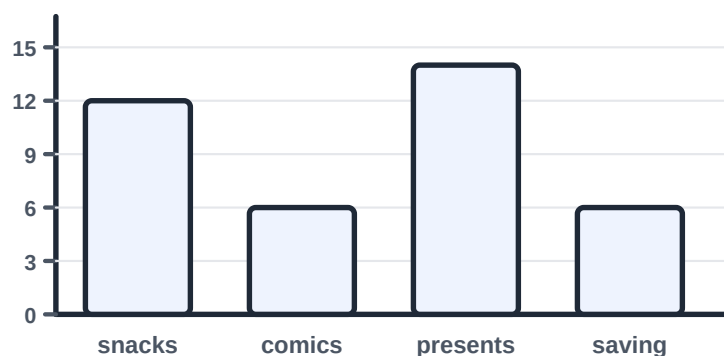
MONEY WARM-UP • 2 MINUTES

Snacks \$12, comics \$6, presents \$14, saving \$6. Largest? _____

WORDS TO KEEP

where it went

The plan sorted by category after the fact, which is a different question from what it came to.



How much, and where

The month came to \$38.00. That is one fact. Where it went is a completely different fact and the plan can only be fixed with the second one.

Drawn as bars, the month answers questions a column of numbers does not. Which is biggest, obviously. But also: are two of them nearly the same? Is one of them much bigger than all the others? Does the shape look like what you thought you were doing with your money?

That last question is the one worth asking out loud. Most people are surprised by their own bars the first time, and the surprise is the useful part.

Read your own month

	Question	Your answer
a	Which category was biggest?	
b	Which was smallest?	
c	Are any two of them close to the same?	
d	Which surprised you?	
e	Does the biggest bar match what you care about most?	

Now draw the bars for what you **planned**, next to the bars for what happened. Which pair of bars is furthest apart?

► ADD TO YOUR PLAN BOOK

Add a page headed Where It Went, with your month drawn twice: planned bars and actual bars, side by side, same scale.

TALK ABOUT IT

Show the two sets of bars to somebody at home and let them ask one question. Would you want somebody else to see your bars? Why or why not?

**STUCK? TRY THIS**

One bar per category. Same scale for both sets, or the comparison is not a comparison.

★ CHALLENGE ZONE

Two people spend the same total in a month and their bars look nothing alike. Is one of them doing better? Say what you would need to know.

Was the Plan Right?

Not: did you keep it. Whether the plan itself was the right plan.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP · 2 MINUTES

Planned \$36.00, spent \$38.00. Over by? _____

What for	Planned	Actual	Over or under
snacks	\$8.00	\$12.00	\$4.00 over
comics	\$8.00	\$6.00	\$2.00 under
presents	\$12.00	\$14.00	\$2.00 over
saving	\$8.00	\$6.00	\$2.00 under
Altogether	\$36.00		

Two different questions people mix up

Did you keep the plan? Mostly not. Was the plan right? Also mostly not, and that is the more useful thing to find out.

Snacks was over by \$4.00 across the month. Every week. That is not four failures of willpower, it is one wrong number: snacks needed \$12.00 and was given \$8.00. A category that is over every single week was measured wrong.

Compare that with a category that was exact three weeks and \$6.00 over once. That one is not a wrong number, that is one event, and the fix for it is the set-aside rather than a bigger category.

Wrong number, or one event?

	The pattern across four weeks	Wrong number or one event?	What you would change
a	over by \$1 every week	wrong number	raise it by \$1
b	exact, exact, exact, \$6 over		
c	under by \$2 every week		
d	over, under, over, under, ends level		
e	over by more each week than the week before		

Row d ends the month at exactly zero difference. Is that plan right? Say what is going on in it.

TALK ABOUT IT

Ask somebody at home whether their plan was wrong or their week was unusual.
How many weeks do you need before you can say a category is the wrong size?

**STUCK? TRY THIS**

Look down the difference column for one category across four weeks. The pattern matters more than any one number.

★ CHALLENGE ZONE

A category is over by \$1, \$2, \$3 and \$4 in four weeks. What is happening, and what will week 5 be? Is raising the amount the right fix?

Change One Thing

One category, one reason, written down. Changing everything teaches you nothing.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP • 2 MINUTES

Snacks goes from \$8 to \$12. Something must fall by? _____

WORDS TO KEEP

change one thing Revise one category and say why. Changing everything teaches you nothing.

What for	Planned
snacks	\$8.00
comics	\$8.00
presents	\$12.00
saving	\$8.00
Altogether	\$36.00

One change, and the money has to come from somewhere

Snacks needs \$12.00 instead of \$8.00. The month still has \$36.00 in it. So \$4.00 has to come out of another category, and you have to say which.

Change one thing at a time. If you change four categories at once and next month is better, you have no idea which change did it, and you have to run the whole month again to find out.

That is the same idea the Grade 5 entrepreneurship students meet with their two selling sessions, and it is worth meeting here first, with your own money, where the only person the experiment is about is you.

Make the change balance

	Category raised	By	Category lowered	By	Does the total hold?
a	snacks	\$4.00	comics	\$4.00	yes
b	presents	\$3.00	saving		
c	saving	\$2.00	snacks		
d	comics	\$5.00	presents		
e	snacks	\$4.00	saving		

Row e takes it out of saving. That balances and it is not the same kind of change as the others. Say why in one sentence.

► ADD TO YOUR PLAN BOOK

Add a page headed One Change, with the category, the amount, where the money came from, and the reason.

TALK ABOUT IT

Make one change to your plan and say the reason out loud to somebody at home.
When is it right to change more than one thing at once?

**STUCK? TRY THIS**

Whatever you add to one category, take from another. The total at the bottom does not move.

★ CHALLENGE ZONE

You want to raise snacks by \$4 and you refuse to lower saving. What are your options? Is doing nothing one of them?

Months, Not Weeks

A goal far enough away that the unit changes, and the plan has to survive it.

IN CLASS THIS WEEK

Multi-step word problems, remainders (4.OA.3, 4.NBT.6)

MONEY WARM-UP • 2 MINUTES

\$96.00 at \$8.00 a month. How many months? _____

WORDS TO KEEP

months, not weeks A goal so far off that the unit changes.

\$8.00 every month



12 of them

\$96.00

Twelve months is not twelve weeks

\$96.00, saving \$8.00 a month, is twelve months. A year.

Grade 3 counted in weeks and the answers came out in numbers you could hold: eight weeks, twenty-four weeks. This is a different size of thing and it deserves a different kind of respect. Twelve months means your birthday happens in the middle of it.

Which is exactly why the arithmetic matters. A goal that takes a year is not a goal you can keep in your head. Written down, with the months ticked off, it is possible. Not written down, it stops in about March, and nobody ever decides to stop.

How long, really?

	The goal	Saved each month	Months	Is that more than a year?
a	\$96.00	\$8.00	12	no, exactly a year
b	\$60.00	\$5.00		
c	\$150.00	\$10.00		
d	\$45.00	\$4.00		
e	\$200.00	\$25.00		

One row does not divide evenly. How many whole months, how much short, and what would you do in the last month?

► ADD TO YOUR PLAN BOOK

Add a page headed The Long One, with a goal, the monthly amount, the number of months, and twelve boxes to tick off.

TALK ABOUT IT

Name one thing you would want in a year. Work out the monthly amount.

What could go wrong over twelve months that would not go wrong over four weeks?

**STUCK? TRY THIS**

Divide the goal by the monthly amount. If it does not divide evenly, you need one more month than the whole number.

★ CHALLENGE ZONE

A goal is \$150 and you can save \$10 a month, but twice a year something takes \$20 out of it. How long does it really take?

Your Plan Book, Finished

Every page of it written by you, about your own money, over a whole school year.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP • 2 MINUTES

A plan of \$9.00 a week, for 36 weeks. How much passed through it? _____

What for	Planned	Actual	Over or under

What is in it

Go through the Plan Book from the front. It should be in order and every page should have your own numbers on it.

The first plan, written in week 5, before you had ever kept one. Your categories. The first plan column. The first month of both columns. The page where something broke and the plan had to eat it. The set-aside. The month drawn as bars, twice. The one change, with its reason.

Nobody else could have written it, which is the point, and it is also why it is worth keeping rather than handing in.

Check the book

	Question	Your answer
a	How many pages are in your Plan Book?	
b	Which page was hardest to fill in honestly?	
c	How much money passed through your plan this year?	
d	How many times did you change the plan?	
e	Which page would you show somebody first?	

If a page is missing, fill it in now from the workbook page it came from. A book with a gap in it is still a book; a book you gave up on in March is not.

► ADD TO YOUR PLAN BOOK

This is the last page. Number every page in the book, front to back, and write the total on this one.

TALK ABOUT IT

Show the whole Plan Book to somebody at home, front to back.

Which page would you keep if you could only keep one?

**STUCK? TRY THIS**

Work from the front. Every page has a heading that came from a week, so any gap has a workbook page waiting to fill it.

★ CHALLENGE ZONE

If you kept this book for another year, what would you add to it that is not in it now?

Decide It Before

The whole year in three words, and what happens to them next year.

IN CLASS THIS WEEK

Multi-step word problems (4.OA.3)

MONEY WARM-UP • 2 MINUTES

You planned \$36.00 and spent \$34.50. Under by? _____

WORDS TO KEEP

decide it before The whole year in three words.



Three words, and why they are in that order

Decide it before. That is the year.

Last year you learned that choosing one thing means not having the other, and that this is true whether or not you notice. This year you moved the noticing earlier, to a table, before the shop, where it is easier and nobody is waiting.

None of it works without the second column. A plan with no record beside it is a plan that is always right, and a plan that is always right has never told anybody anything. The checking is not the boring part after the planning. It is the part that makes the planning worth doing.

The last questions

	Question	Your answer
a	What is the one thing you would tell a younger student?	
b	What did your plan say about you that surprised you?	
c	Which was harder, making a plan or keeping the record?	
d	What would you plan for next, that you did not this year?	
e	Will you keep doing this? Say honestly.	

Row e is allowed to be no. If it is, say what would have to be different for it to be yes, and that is a more useful answer than a polite one.

 TALK ABOUT IT

Tell somebody at home the three words and what they mean.

What is one thing you now do differently with money?

**STUCK? TRY THIS**

There is no arithmetic on this page. Answer in sentences and take your time over row b.

★ CHALLENGE ZONE

Write the plan you would make for a whole year, in months rather than weeks. What is different about it, apart from the numbers?



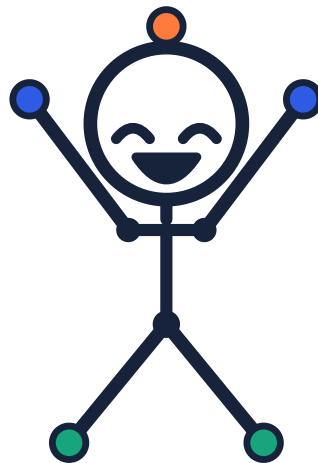
Young Innovators
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Grade 4 · The Planning Year

Certificate of Completion

This is to certify that

completed thirty-six weeks of planning: counted money to the cent, wrote a plan before spending rather than after, kept the record beside it week after week, found out where the plan was wrong and changed one thing at a time, set money aside for what could not be predicted, and finished the year able to say what their own money actually does.



Teacher

Date

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